RAJALAKSHMI ENGINEERING COLLEGE

Nevash S 251001162-2025-IT

N2

Started on	Thursday, 17 September 2026, 10:15 AM
State	Finished
Completed on	Thursday, 17 September 2026, 10:23 AM
Time taken	7 mins 32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int binarySearch(int a[], int low, int high, int x) {
    if (low > high)
        return -1;
    int mid = (low + high) / 2;
    if (a[mid] == x)
        return mid;
    if (a[mid] > x)
        return binarySearch(a, low, mid - 1, x);
    return binarySearch(a, mid + 1, high, x);
}
int main() {
    int n, x, i, pos;
    scanf("%d", &n);
    int a[n];
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &x);
    for (i = 0; i < n; i++) {
        pos = binarySearch(a, i + 1, n - 1, x - a[i]);
        if (pos != -1) {
            printf("%d\n%d", a[i], a[pos]);
            return 0;
        }
    }
    printf("No");
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔

	Input	Expected	Got	
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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